Amirjon Abdunayimov

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EDUCATION:

New York University — *Tandon School of Engineering* Expected Graduation: May 2029

Bachelor of Science in Computer Science

Academic Honors (Kingsborough CC): Jack Kent Cooke Transfer Scholarship Semifinalist, Honors Scholar, Phi Theta Kappa, Dean's List

WORK EXPERIENCE:

Boardwalk Labs — *Software Engineering Intern (Part-Time)* June 2026 – Sep 2026

- Built and optimized **Go/Python microservices** for OctoPilot AI, reducing **end-to-end processing latency by 50%** through background worker queues and connection pooling.
- Co-developed **Go credit-accounting APIs** with per-user balance tracking and **idempotency keys**, preventing duplicate credit deductions under request retries.
- Engineered an **LLM agent execution pipeline** with **Redis TTL caching**, **token-bucket rate limiting**, and **Prometheus instrumentation** to control request volume and detect streaming failures.

Sound of Earth — *DevOps & Cloud Infrastructure Intern (Part-Time)* May 2026 – Aug 2026

- Built and maintained **CI/CD pipelines with Docker, GitHub Actions, and Linux**, automating deployments across environments to reduce production regressions by **40%** and staging costs by **$3.2K/year**.
- Built a zero-downtime Supabase/PostgreSQL migration pipeline for **138,000+ records**, organizing legacy data into **8 normalized tables** and developing **PostgREST-backed APIs** for web and mobile synchronization.
- Identified and fixed **29 security issues** across production infrastructure, strengthening JWT authentication, CORS configuration, secrets management, and Docker network isolation.

Kingsborough Community College Learning Center — *Computer Science & Mathematics Tutor* May 2026 – Aug 2026

- Tutored students in C++, algebra, pre-calculus, and statistics, using code tracing, debugging exercises, and worked examples to strengthen independent problem-solving skills.

PROJECTS:

GhostMirror — Developer Activity Dashboard | React, TypeScript, Python, FastAPI, SQLite

- Built a **React and TypeScript** dashboard connected to FastAPI services for capturing, searching, editing, and reviewing developer activity stored in SQLite.
- Implemented typed API integration and query caching with TanStack Query, supporting event filters, activity summaries, monitor controls, and JSON import/export.
- Added backend tests with pytest, component tests with Vitest and React Testing Library, and browser smoke tests with Playwright; integrated automated checks into GitHub Actions.

sev0 — Autonomous AI Incident Investigation & Repair System | Python, Anthropic API, FastAPI, PostgreSQL, Docker, Prometheus, Loki

- Built an **autonomous AI software engineer** for production incidents, orchestrating **16 diagnostic tools** across Prometheus, Loki, Git, and Docker to trace failures to the root-cause service, file, symbol, and introducing commit.
- Designed a **4-service incident lab with 23 fault scenarios**, including **35%** adversarial and silent failures, to evaluate root-cause analysis across code, configuration, infrastructure, and business-metric regressions.
- Engineered a **sandboxed repair-and-verification pipeline** that reproduces failures before patching, runs regression checks on proposed fixes, enforces patch safety limits, and generates **draft GitHub PRs** for human review.

LEADERSHIP:

AI, Coding, and Systems (ACS) Club (Kingsborough CC) — *Founder & President* March 2026 – June 2026

- Founded a computer science organization focused on **software engineering, AI, and systems programming**, leading technical workshops, project-based learning, and peer development sessions.

PROFESSIONAL SKILLS:

Programming Languages: Python, TypeScript, JavaScript, SQL, Go, C++, Dart, HTML/CSS
Frameworks & Databases: React, FastAPI, Node.js, Flutter, PostgreSQL, SQLite, Redis
Tools & Infrastructure: Docker, Linux, Git, GitHub Actions (CI/CD), REST APIs, WebSockets, Prometheus
Testing: pytest, Vitest, React Testing Library, Playwright
Microsoft Office: Excel, Word, PowerPoint